MIL-S-36653
20 March 1970

MILITARY SPECIFICATION

SCREW, BONE, 82° COUNTERSUNK HEAD, PILOT POINT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers screws used in the internal fixation of bone plates to the human bone.

1.2 Classification. Bone screws shall be of the following sizes as specified below (see 6.1):

Size	*Length (in inches excluding screw head)
1	1/4
2	3/8
3	1/2
4	5/8
5	3/4
6	7/8
7	1
8	1-1/8
9	1-1/4
10	1-3/8
11	1-1/2
12	1-5/8
13	1-3/4
14	1-7/8
15	2

*Length of screw shall designate effective length of threads.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood,
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWING

Defense Personnel Support Center (DPSC)

24086 SCREW, BONE

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F55-66	Stainless Steel Bars and Wire For Surgical Implants
ASTM F86-68	Recommended Practice for Surface Requirements of Metallic Surgical Implants

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I. B1.1	Unified Screw Threads
A.N.S.I. B1.2	Gages and Gaging for Unified Screw Threads

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

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3. REQUIREMENTS

3.1 Material. The bone screws shall be fabricated from cold drawn wire conforming to the composition specified in ASTM F55-66 concerning stainless steel material used for surgical implants, except that the carbon content shall be 0.03 percent maximum. Tensile strength of wire shall be 136,000 pounds per square inch (psi), minimum when tested as specified in 4.4.2.

3.1.1 Inclusion content of material. The nonmetallic inclusion content of the material used in the construction of the bone screws shall be determined by the JERKONTORET (JK) Chart method in accordance with ASTM E45-63, using methods "A" or "D". The numbers listed in Table I shall represent the maximum allowable inclusion ratings.

Table I. Maximum Allowable Inclusion Ratings In Accordance with ASTM E45

	Type A	Type B	Type C	Type D
Thin Series	2	1-1/2	1-1/2	2
Thick Series	1-1/2	1	1	1

3.2 Hardness. After proper stress relieving, Rockwell hardness of the bone screws shall be C30 to C35 when tested as specified in 4.4.3.

3.3 Finish. Heads of bone screws shall have a satin finish. Screws shall not be electropolished. Definitions of acceptable surface characteristics shall be as described in ASTM F86.

3.3.1 Passivation. The bone screws shall be properly passivated after completion of all final finishing operations and shall be accomplished by the methods described in ASTM F86.

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3.4 Style, design and dimensions. Bone screws shall be suitable for implanting various bone plates and hip nails into the body during surgical procedures involving traction reduction of fractures and internal fixation of the human bone. Screws shall have cruciate slotted heads, milled concave suitable for use with FSN 6515-721-9835, SCREWDRIVER, BONE SCREW. Screws shall have pilot points, for insertion ease, and a 45° relief on the lead tap thread of the screw as shown on drawing. The threads of the screw shall be in accordance with A.N.S.I. Standard B1.1 for 6-32 UNC-3A (Unified Screw Thread System of Dimensions). The threaded portion of the screw shall extend to and approximate the head of the screw. Slots in screw head shall not extend into the side wall of the head (i.e., shall not extend to the outermost surface of the head). Bone screws shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing No. 24086 for appropriate size, item identification and stock number.

3.5 External threads of the screw. External threads of the screw shall allow the "GO" thread ring gages to pass over their full length and turn freely and shall not allow passage of the "LO" and "NOT GO" thread ring gages when tested as specified in 4.4.4. Definitions of acceptance and rejection of the gaging tests shall be as specified in A.N.S.I. B1.2.

3.6 Resistance to boil test, copper sulfate test and saline test. Bone screws shall show no evidence of corrosion when subjected to the boil test as specified in 4.4.1.1, the copper sulfate test as specified in 4.4.1.2, and the saline test as specified in 4.4.1.3.

3.7 Workmanship. Workmanship shall be first class throughout. Screw threads and screw head slots shall be free from burrs, pits and tool marks. Bone screws shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II.

Table II. Sampling Criteria for Examination

Type Defect	Inspection Level*	AQL (percent defective)
Major	S-4	1.0
Minor	S-2	2.5

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III.

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Table III. Sampling Criteria for Tests

Component	Characteristic	Req't	Test Procedure	Unit of Product	Insp. Level	AQL
Bone Screw	Dimensions	3.4	- - -	1-screw	S-2	1.5
	Tensile strength	3.1	4.4.2	1-screw	S-2	1.5
	Inclusion Content	3.1.1	ASTM E45	1-screw *	S-1	1.0
	Hardness	3.2	4.4.3	1-screw	S-2	1.5
	External Thread Test	3.5	4.4.4	1-screw	S-4	0.65
	Boil Test	3.6	4.4.1.1	1-screw	S-4	0.65
	Copper Sulfate Test	3.6	4.4.1.2	1-screw	S-4	0.65
	Saline Test	3.6	4.4.1.3	1-screw	S-4	0.65

* The unit lot of steel, for which the inclusion content is to be determined, shall not be larger than one heat.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the contraction of the bone screws shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bone screws shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Inspection, however, shall not be restricted to defects listed in Table IV. Any bone screws in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Bone screws not of finish specified.
102	Heads of screws not of cruciated design.
103	Screws do not have pilot points.
104	Threads of screw not complete or well defined or do not extend to screw head.
105	Slots in screw head extend out to the most outer surface of head.
106	Screw not free from burrs, pits, cracks and tool marks.
Minor	
201	Screws not properly protected and properly identified in unit package.

4.4 Tests.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Bone screws shall be subjected to the boil test as described and specified in Federal Specification GG-I-526.

4.4.1.2 Copper Sulfate Test. Bone screws shall be subjected to the copper sulfate test as described in GG-I-526.

4.4.1.3 Saline test. Bone screws shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Bone screws shall then be completely immersed in Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 $\pm$ 1 degrees Centigrade (^{o}C) for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Tensile strength test. Tensile strength of the bone screws shall be conducted in accordance with Test Method 211.1 of Federal Test Method Standard No. 151.

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4.4.3 Hardness test. Rockwell hardness of the bone screws shall be determined in accordance with method 243 of Federal Test Method Standard No. 151.

4.4.4 External thread test. The contractor shall provide all of the following thread ring gages to the Government Quality Assurance Representative (QAR) for checking the 6-32 UNC-3A external threads of the bone screw with respect to Pitch Diameter and Major Diameter.

a. "GO" Thread Ring Gage for PITCH DIAMETER (X gage)
Identification marking: 6-32 UNC-3A GO PD .1177

b. "LO" Thread Ring Gage for PITCH DIAMETER (X gage)
Identification marking: 6-32 UNC-3A LO PD .1156

c. "GO" Plain Ring Gage for MAJOR DIAMETER (Z gage)
Identification marking: 6-32 UNC 3A GO .1380

d. "NOT GO" Plain Ring Gage for MAJOR DIAMETER (Z gage)
Identification marking: 6-32 UNC 3A NOT GO .1320

The gaging procedure to be used, dimensions, tolerances and gage identifications shall be in accordance with A.N.S.I. B1.2. Fits of gages onto threads of screws shall be noted.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One screw, of the stock number specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
6 units	144 units	3,456 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Six screws (6 units) of like stock number shall be individually wrapped in tissue, or shall be mounted on a card with suitable die-cuts to prevent contact of screws with each other. Six screws, so prepared, shall be further packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening. Each envelope shall be further packaged in accordance with MIL-P-116, method 1A-8.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

1. Title, number and date of this specification.
2. Applicable size (see 6.2).
3. Applicable levels of packaging and packing (see 5.1 and 5.2).
4. Federal stock number (see 6.2).

6.2 The items covered by this specification are listed in the Federal Supply Catalog as follows:

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Federal Stock No.	Item Identification
6515-366-3010	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1/4 inch, Effective Thread Length
6515-366-3015	SCREW, BONE, 82° Countersunk Head, Pilot Point, 3/8 inch, Effective Thread Length
6515-366-3020	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1/2 inch, Effective Thread Length
6515-366-3025	SCREW, BONE, 82° Countersunk Head, Pilot Point, 5/8 inch, Effective Thread Length
6515-366-3030	SCREW, BONE, 82° Countersunk Head, Pilot Point, 3/4 inch, Effective Thread Length
6515-366-3035	SCREW, BONE, 82° Countersunk Head, Pilot Point, 7/8 inch, Effective Thread Length
6515-366-3040	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1 inch, Effective Thread Length
6515-366-3045	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-1/8 inch, Effective Thread Length
6515-366-3050	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-1/4 inch, Effective Thread Length
6515-366-3055	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-3/8 inch, Effective Thread Length
6515-366-3060	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-1/2 inch, Effective Thread Length
6515-366-3065	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-5/8 inch, Effective Thread Length
6515-366-3070	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-3/4 inch, Effective Thread Length

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Federal Stock No.	Item Identification
6515-366-3075	SCREW, BONE, 82° Countersunk Head, Pilot Point, 1-7/8 inch, Effective Thread Length
6515-366-3080	SCREW, BONE, 82° Countersunk Head, Pilot Point, 2 inch, Effective Thread Length

6.3 This specification is intended to cover the types of bone screws which are normally procured to meet Military requirements. It does not include all types or sizes as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes requirements of Purchase Description No. 8, dated 24 November 1967 and supersedes requirements for Bone Screws as indicated in Military Specification MIL-B-16505 (BuMed), date 15 July 1951.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MS
Navy - MD
Air Force - 03

Preparing activity:
Defense Supply Agency - DM
Project No. 6515-1461

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: MIL-S-36653 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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